

```

=====
FIVE-SEED EXPERIMENT
=====
Seeds: [42, 52, 62, 72, 82]
Epochs: 10
Batch size: 64
Learning rate: 0.0001
Mounted at /content/drive

=====
DATASET PATHS
=====
Train directory: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/train
Validation directory: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/val
Test directory: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/test

=====
DATASET INFORMATION
=====
Classes: ['fractured', 'not fractured']
Training images: 9246
Validation images: 829
Testing images: 506

=====
DEVICE
=====
Using device: cuda
GPU: Tesla T4

Results will be permanently saved to:
/content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results

=====
STARTING FIVE-SEED EXPERIMENT
Seeds: [42, 52, 62, 72, 82]
=====

#####
RUNNING SEED 42 (1/5)
#####

=====
STARTING SEED 42
=====

Random Seed: 42
Seed 42: creating DataLoaders...
Seed 42: DataLoaders ready | Train batches=145, Val batches=13, Test batches=8
Seed 42: creating EfficientNet-B0 model...
Downloading: "https://download.pytorch.org/models/efficientnet_b0_rwightman-7f5810bc.pth" to /root/.cache/torch/hub/checkpoints/efficientnet_b0.
100%|██████████| 20.5M/20.5M [00:00<00:00, 216MB/s]
Seed 42: model created successfully.

No checkpoint found for Seed 42.
Starting Seed 42 from Epoch 1.

=====
SEED 42 | EPOCH 1/10 | STARTING TRAINING
=====

Seed 42 | Epoch 1/10: 100% 145/145 [53:33<00:00, 18.25s/batch, acc=66.22%, loss=0.6354]
Seed 42 | Epoch 1: FIRST BATCH COMPLETED in 46.74 seconds | Loss=0.6935

Seed 42 | Epoch 1: training finished. Starting validation...
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 42 | Epoch 1: validation finished.

-----
Seed 42 | Epoch 1/10
Train Loss: 0.6354 | Val Loss: 0.5757
Train Accuracy: 66.22% | Val Accuracy: 91.44%
Learning Rate: 0.00010000
Epoch Time: 63.33 minutes
Total Seed Time: 63.34 minutes
Best Val Loss: 0.5757 | Best Model: UPDATED
-----
Checkpoint saved successfully: Seed 42, Epoch 1
Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_42_checkpoint

```

```
=====
SEED 42 | EPOCH 2/10 | STARTING TRAINING
=====
Seed 42 | Epoch 2/10:  0%|          | 0/145 [00:00<?, ?batch/s]
Seed 42 | Epoch 2: FIRST BATCH COMPLETED in 0.42 seconds | Loss=0.5875

Seed 42 | Epoch 2: training finished. Starting validation...
Seed 42 | Epoch 2: validation finished.
```

```
-----
Seed 42 | Epoch 2/10
Train Loss: 0.5607 | Val Loss: 0.5431
Train Accuracy: 97.85% | Val Accuracy: 95.54%
Learning Rate: 0.00010000
Epoch Time: 1.24 minutes
Total Seed Time: 64.59 minutes
Best Val Loss: 0.5431 | Best Model: UPDATED
-----
Checkpoint saved successfully: Seed 42, Epoch 2
Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_42_checkpoint
```

```
=====
SEED 42 | EPOCH 3/10 | STARTING TRAINING
=====
Seed 42 | Epoch 3/10:  0%|          | 0/145 [00:00<?, ?batch/s]
Seed 42 | Epoch 3: FIRST BATCH COMPLETED in 0.58 seconds | Loss=0.5579

Seed 42 | Epoch 3: training finished. Starting validation...
Seed 42 | Epoch 3: validation finished.
```

```
-----
Seed 42 | Epoch 3/10
Train Loss: 0.5310 | Val Loss: 0.5188
Train Accuracy: 98.98% | Val Accuracy: 96.86%
Learning Rate: 0.00001000
Epoch Time: 1.26 minutes
Total Seed Time: 65.86 minutes
Best Val Loss: 0.5188 | Best Model: UPDATED
-----
Checkpoint saved successfully: Seed 42, Epoch 3
Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_42_checkpoint
```

```
=====
SEED 42 | EPOCH 4/10 | STARTING TRAINING
=====
Seed 42 | Epoch 4/10:  0%|          | 0/145 [00:00<?, ?batch/s]
Seed 42 | Epoch 4: FIRST BATCH COMPLETED in 0.52 seconds | Loss=0.5129

Seed 42 | Epoch 4: training finished. Starting validation...
Seed 42 | Epoch 4: validation finished.
```

```
-----
Seed 42 | Epoch 4/10
Train Loss: 0.5157 | Val Loss: 0.5159
Train Accuracy: 99.21% | Val Accuracy: 96.14%
Learning Rate: 0.00001000
Epoch Time: 1.26 minutes
Total Seed Time: 67.13 minutes
Best Val Loss: 0.5159 | Best Model: UPDATED
-----
Checkpoint saved successfully: Seed 42, Epoch 4
Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_42_checkpoint
```

```
=====
SEED 42 | EPOCH 5/10 | STARTING TRAINING
=====
Seed 42 | Epoch 5/10:  0%|          | 0/145 [00:00<?, ?batch/s]
Seed 42 | Epoch 5: FIRST BATCH COMPLETED in 0.44 seconds | Loss=0.5126

Seed 42 | Epoch 5: training finished. Starting validation...
Seed 42 | Epoch 5: validation finished.
```

```
-----
Seed 42 | Epoch 5/10
Train Loss: 0.5122 | Val Loss: 0.5131
Train Accuracy: 99.46% | Val Accuracy: 96.38%
Learning Rate: 0.00001000
Epoch Time: 1.25 minutes
```

Total Seed Time: 68.38 minutes
Best Val Loss: 0.5131 | Best Model: UPDATED

Checkpoint saved successfully: Seed 42, Epoch 5

Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_42_checkpoint

=====

SEED 42 | EPOCH 6/10 | STARTING TRAINING

=====

Seed 42 | Epoch 6/10: 0%| | 0/145 [00:00<?, ?batch/s]
Seed 42 | Epoch 6: FIRST BATCH COMPLETED in 0.43 seconds | Loss=0.5074

Seed 42 | Epoch 6: training finished. Starting validation...
Seed 42 | Epoch 6: validation finished.

Seed 42 | Epoch 6/10
Train Loss: 0.5095 | Val Loss: 0.5105
Train Accuracy: 99.50% | Val Accuracy: 96.02%
Learning Rate: 0.00000100
Epoch Time: 1.28 minutes
Total Seed Time: 69.67 minutes
Best Val Loss: 0.5105 | Best Model: UPDATED

Checkpoint saved successfully: Seed 42, Epoch 6

Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_42_checkpoint

=====

SEED 42 | EPOCH 7/10 | STARTING TRAINING

=====

Seed 42 | Epoch 7/10: 0%| | 0/145 [00:00<?, ?batch/s]
Seed 42 | Epoch 7: FIRST BATCH COMPLETED in 0.58 seconds | Loss=0.5083

Seed 42 | Epoch 7: training finished. Starting validation...
Seed 42 | Epoch 7: validation finished.

Seed 42 | Epoch 7/10
Train Loss: 0.5086 | Val Loss: 0.5110
Train Accuracy: 99.37% | Val Accuracy: 96.14%
Learning Rate: 0.00000100
Epoch Time: 1.26 minutes
Total Seed Time: 70.93 minutes
Best Val Loss: 0.5105 | Best Model: NOT UPDATED

Checkpoint saved successfully: Seed 42, Epoch 7

Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_42_checkpoint

=====

SEED 42 | EPOCH 8/10 | STARTING TRAINING

=====

Seed 42 | Epoch 8/10: 0%| | 0/145 [00:00<?, ?batch/s]
Seed 42 | Epoch 8: FIRST BATCH COMPLETED in 0.64 seconds | Loss=0.5025

Seed 42 | Epoch 8: training finished. Starting validation...
Seed 42 | Epoch 8: validation finished.

Seed 42 | Epoch 8/10
Train Loss: 0.5080 | Val Loss: 0.5109
Train Accuracy: 99.43% | Val Accuracy: 96.02%
Learning Rate: 0.00000100
Epoch Time: 1.26 minutes
Total Seed Time: 72.20 minutes
Best Val Loss: 0.5105 | Best Model: NOT UPDATED

Checkpoint saved successfully: Seed 42, Epoch 8

Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_42_checkpoint

=====

SEED 42 | EPOCH 9/10 | STARTING TRAINING

=====

Seed 42 | Epoch 9/10: 0%| | 0/145 [00:00<?, ?batch/s]
Seed 42 | Epoch 9: FIRST BATCH COMPLETED in 0.48 seconds | Loss=0.5194

Seed 42 | Epoch 9: training finished. Starting validation...
Seed 42 | Epoch 9: validation finished.

```
-----
Seed 42 | Epoch 9/10
Train Loss: 0.5075 | Val Loss: 0.5102
Train Accuracy: 99.50% | Val Accuracy: 96.62%
Learning Rate: 0.00000010
Epoch Time: 1.26 minutes
Total Seed Time: 73.46 minutes
Best Val Loss: 0.5102 | Best Model: UPDATED
-----
```

```
Checkpoint saved successfully: Seed 42, Epoch 9
Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_42_checkpoint_9.pth
```

```
=====
SEED 42 | EPOCH 10/10 | STARTING TRAINING
=====
```

```
Seed 42 | Epoch 10/10: 0% | 0/145 [00:00<?, ?batch/s]
Seed 42 | Epoch 10: FIRST BATCH COMPLETED in 0.38 seconds | Loss=0.4975
```

```
Seed 42 | Epoch 10: training finished. Starting validation...
Seed 42 | Epoch 10: validation finished.
```

```
-----
Seed 42 | Epoch 10/10
Train Loss: 0.5078 | Val Loss: 0.5095
Train Accuracy: 99.45% | Val Accuracy: 97.35%
Learning Rate: 0.00000010
Epoch Time: 1.25 minutes
Total Seed Time: 74.72 minutes
Best Val Loss: 0.5095 | Best Model: UPDATED
-----
```

```
Checkpoint saved successfully: Seed 42, Epoch 10
Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_42_checkpoint_10.pth
```

```
=====
SEED 42 TRAINING COMPLETE
=====
```

```
Total time: 74.73 minutes
Best validation-loss model loaded for Seed 42.
```

```
Seed 42: starting final training-set evaluation...
Seed 42: training-set evaluation complete.
Seed 42: starting validation-set evaluation...
Seed 42: validation-set evaluation complete.
Seed 42: starting testing-set evaluation...
Seed 42: testing-set evaluation complete.
```

```
=====
SEED 42 - TRAINING RESULTS
=====
```

```
Accuracy (%): 99.6864
Sensitivity (%): 99.5690
Specificity (%): 99.8046
Precision (%): 99.8056
F1-Score (%): 99.6871
Error Rate (%): 0.3136
ROC-AUC: 0.9996
```

```
=====
SEED 42 - VALIDATION RESULTS
=====
```

```
Accuracy (%): 97.3462
Sensitivity (%): 96.7480
Specificity (%): 98.2196
Precision (%): 98.7552
F1-Score (%): 97.7413
Error Rate (%): 2.6538
ROC-AUC: 0.9947
```

```
=====
SEED 42 - TESTING RESULTS
=====
```

```
Accuracy (%): 96.0474
Sensitivity (%): 94.0299
Specificity (%): 98.3193
Precision (%): 98.4375
F1-Score (%): 96.1832
Error Rate (%): 3.9526
ROC-AUC: 0.9928
```



```
07/09/2026, 18:38                                output EFFICIENTNET-B0 + NONLINEAR CLASSICAL MLP.ipynb - Colab

=====
SEED 42 - TRAINING CONFUSION MATRIX
=====
[[4597   9]
 [ 20 4620]]

=====
SEED 42 - VALIDATION CONFUSION MATRIX
=====
[[331   6]
 [ 16 476]]

=====
SEED 42 - TESTING CONFUSION MATRIX
=====
[[234   4]
 [ 16 252]]

=====
SEED 42 PERFORMANCE COMPARISON
=====
      Metric Training Validation Testing
Accuracy (%) 99.686351  97.346200 96.047431
Sensitivity (%) 99.568966 96.747967 94.029851
Specificity (%) 99.804603 98.219585 98.319328
Precision (%) 99.805574 98.755187 98.437500
F1-Score (%) 99.687129 97.741273 96.183206
Error Rate (%) 0.313649  2.653800  3.952569
ROC-AUC 0.999610  0.994741  0.992788

Seed 42 completion marker saved.
Seed 42 result saved: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/seed_results/seed_42_re

=====
FINISHED SEED 42
=====
Completed seeds: [42]
Remaining seeds: [52, 62, 72, 82]

#####
RUNNING SEED 52 (2/5)
#####

=====
STARTING SEED 52
=====

Random Seed: 52
Seed 52: creating DataLoaders...
Seed 52: DataLoaders ready | Train batches=145, Val batches=13, Test batches=8
Seed 52: creating EfficientNet-B0 model...
Seed 52: model created successfully.

No checkpoint found for Seed 52.
Starting Seed 52 from Epoch 1.

=====
SEED 52 | EPOCH 1/10 | STARTING TRAINING
=====

Seed 52 | Epoch 1/10: 100%                                                    145/145 [01:02<00:00, 2.53batch/s, acc=50.18%, loss=0.6103]
Seed 52 | Epoch 1: FIRST BATCH COMPLETED in 0.53 seconds | Loss=0.6875

Seed 52 | Epoch 1: training finished. Starting validation...
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 52 | Epoch 1: validation finished.

-----
Seed 52 | Epoch 1/10
Train Loss: 0.6103 | Val Loss: 0.5249
Train Accuracy: 50.18% | Val Accuracy: 59.35%
Learning Rate: 0.00010000
Epoch Time: 1.20 minutes
Total Seed Time: 1.20 minutes
Best Val Loss: 0.5249 | Best Model: UPDATED
-----
Checkpoint saved successfully: Seed 52, Epoch 1

https://colab.research.google.com/drive/1tS41tZLdVdxNQxhtPkfVJjRNRINUkQnt
```

Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_52_checkpoint

=====

SEED 52 | EPOCH 2/10 | STARTING TRAINING

=====

Seed 52 | Epoch 2/10: 100% 145/145 [01:01<00:00, 2.65batch/s, acc=50.18%, loss=0.5303]

Seed 52 | Epoch 2: FIRST BATCH COMPLETED in 0.52 seconds | Loss=0.5213

Seed 52 | Epoch 2: training finished. Starting validation...

Seed 52 | Epoch 2: validation finished.

Seed 52 | Epoch 2/10
Train Loss: 0.5303 | Val Loss: 0.4916
Train Accuracy: 50.18% | Val Accuracy: 59.35%
Learning Rate: 0.00010000
Epoch Time: 1.21 minutes
Total Seed Time: 2.42 minutes
Best Val Loss: 0.4916 | Best Model: UPDATED

Checkpoint saved successfully: Seed 52, Epoch 2

Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_52_checkpoint

=====

SEED 52 | EPOCH 3/10 | STARTING TRAINING

=====

Seed 52 | Epoch 3/10: 100% 145/145 [01:03<00:00, 3.01batch/s, acc=84.36%, loss=0.4949]

Seed 52 | Epoch 3: FIRST BATCH COMPLETED in 0.48 seconds | Loss=0.4942

Seed 52 | Epoch 3: training finished. Starting validation...

Seed 52 | Epoch 3: validation finished.

Seed 52 | Epoch 3/10
Train Loss: 0.4949 | Val Loss: 0.4520
Train Accuracy: 84.36% | Val Accuracy: 96.98%
Learning Rate: 0.00001000
Epoch Time: 1.24 minutes
Total Seed Time: 3.66 minutes
Best Val Loss: 0.4520 | Best Model: UPDATED

Checkpoint saved successfully: Seed 52, Epoch 3

Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_52_checkpoint

=====

SEED 52 | EPOCH 4/10 | STARTING TRAINING

=====

Seed 52 | Epoch 4/10: 100% 145/145 [01:04<00:00, 1.94batch/s, acc=98.91%, loss=0.4750]

Seed 52 | Epoch 4: FIRST BATCH COMPLETED in 0.55 seconds | Loss=0.4678

Seed 52 | Epoch 4: training finished. Starting validation...

Seed 52 | Epoch 4: validation finished.

Seed 52 | Epoch 4/10
Train Loss: 0.4750 | Val Loss: 0.4480
Train Accuracy: 98.91% | Val Accuracy: 97.10%
Learning Rate: 0.00001000
Epoch Time: 1.23 minutes
Total Seed Time: 4.89 minutes
Best Val Loss: 0.4480 | Best Model: UPDATED

Checkpoint saved successfully: Seed 52, Epoch 4

Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_52_checkpoint

=====

SEED 52 | EPOCH 5/10 | STARTING TRAINING

=====

Seed 52 | Epoch 5/10: 100% 145/145 [01:04<00:00, 3.03batch/s, acc=99.23%, loss=0.4714]

Seed 52 | Epoch 5: FIRST BATCH COMPLETED in 0.63 seconds | Loss=0.4560

Seed 52 | Epoch 5: training finished. Starting validation...

Seed 52 | Epoch 5: validation finished.

Seed 52 | Epoch 5/10
Train Loss: 0.4714 | Val Loss: 0.4447
Train Accuracy: 99.23% | Val Accuracy: 97.10%
Learning Rate: 0.00001000
Epoch Time: 1.25 minutes
Total Seed Time: 6.15 minutes
Best Val Loss: 0.4447 | Best Model: UPDATED

Checkpoint saved successfully: Seed 52, Epoch 5
Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_52_checkpoint

=====

SEED 52 | EPOCH 6/10 | STARTING TRAINING

=====

Seed 52 | Epoch 6/10: 100% 145/145 [01:03<00:00, 2.88batch/s, acc=99.34%, loss=0.4682]

Seed 52 | Epoch 6: FIRST BATCH COMPLETED in 0.51 seconds | Loss=0.4441

Seed 52 | Epoch 6: training finished. Starting validation...

Seed 52 | Epoch 6: validation finished.

Seed 52 | Epoch 6/10
Train Loss: 0.4682 | Val Loss: 0.4422
Train Accuracy: 99.34% | Val Accuracy: 97.10%
Learning Rate: 0.00001000
Epoch Time: 1.23 minutes
Total Seed Time: 7.39 minutes
Best Val Loss: 0.4422 | Best Model: UPDATED

Checkpoint saved successfully: Seed 52, Epoch 6
Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_52_checkpoint

=====

SEED 52 | EPOCH 7/10 | STARTING TRAINING

=====

Seed 52 | Epoch 7/10: 100% 145/145 [01:04<00:00, 2.34batch/s, acc=99.50%, loss=0.4657]

Seed 52 | Epoch 7: FIRST BATCH COMPLETED in 0.41 seconds | Loss=0.4937

Seed 52 | Epoch 7: training finished. Starting validation...

Seed 52 | Epoch 7: validation finished.

Seed 52 | Epoch 7/10
Train Loss: 0.4657 | Val Loss: 0.4414
Train Accuracy: 99.50% | Val Accuracy: 97.23%
Learning Rate: 0.00001000
Epoch Time: 1.23 minutes
Total Seed Time: 8.63 minutes
Best Val Loss: 0.4414 | Best Model: UPDATED

Checkpoint saved successfully: Seed 52, Epoch 7
Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_52_checkpoint

=====

SEED 52 | EPOCH 8/10 | STARTING TRAINING

=====

Seed 52 | Epoch 8/10: 100% 145/145 [01:04<00:00, 2.97batch/s, acc=99.46%, loss=0.4654]

Seed 52 | Epoch 8: FIRST BATCH COMPLETED in 0.49 seconds | Loss=0.4759

Seed 52 | Epoch 8: training finished. Starting validation...

Seed 52 | Epoch 8: validation finished.

Seed 52 | Epoch 8/10
Train Loss: 0.4654 | Val Loss: 0.4416
Train Accuracy: 99.46% | Val Accuracy: 97.23%
Learning Rate: 0.00001000
Epoch Time: 1.25 minutes
Total Seed Time: 9.88 minutes
Best Val Loss: 0.4414 | Best Model: NOT UPDATED

Checkpoint saved successfully: Seed 52, Epoch 8
Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_52_checkpoint

=====

```
SEED 52 | EPOCH 9/10 | STARTING TRAINING
```

```
Seed 52 | Epoch 9/10: 100%
```

```
145/145 [01:03<00:00, 3.03batch/s, acc=99.37%, loss=0.4653]
```

```
Seed 52 | Epoch 9: FIRST BATCH COMPLETED in 0.46 seconds | Loss=0.4692
```

```
Seed 52 | Epoch 9: training finished. Starting validation...
```

```
Seed 52 | Epoch 9: validation finished.
```

```
Seed 52 | Epoch 9/10
```

```
Train Loss: 0.4653 | Val Loss: 0.4412
```

```
Train Accuracy: 99.37% | Val Accuracy: 97.35%
```

```
Learning Rate: 0.00000010
```

```
Epoch Time: 1.24 minutes
```

```
Total Seed Time: 11.13 minutes
```

```
Best Val Loss: 0.4412 | Best Model: UPDATED
```

```
Checkpoint saved successfully: Seed 52, Epoch 9
```

```
Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_52_checkpoint
```

```
SEED 52 | EPOCH 10/10 | STARTING TRAINING
```

```
Seed 52 | Epoch 10/10: 100%
```

```
145/145 [01:04<00:00, 2.55batch/s, acc=99.47%, loss=0.4653]
```

```
Seed 52 | Epoch 10: FIRST BATCH COMPLETED in 0.55 seconds | Loss=0.4921
```

```
Seed 52 | Epoch 10: training finished. Starting validation...
```

```
Seed 52 | Epoch 10: validation finished.
```

```
Seed 52 | Epoch 10/10
```

```
Train Loss: 0.4653 | Val Loss: 0.4417
```

```
Train Accuracy: 99.47% | Val Accuracy: 97.35%
```

```
Learning Rate: 0.00000010
```

```
Epoch Time: 1.23 minutes
```

```
Total Seed Time: 12.37 minutes
```

```
Best Val Loss: 0.4412 | Best Model: NOT UPDATED
```

```
Checkpoint saved successfully: Seed 52, Epoch 10
```

```
Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_52_checkpoint
```

```
SEED 52 TRAINING COMPLETE
```

```
Total time: 12.37 minutes
```

```
Best validation-loss model loaded for Seed 52.
```

```
Seed 52: starting final training-set evaluation...
```

```
Seed 52: training-set evaluation complete.
```

```
Seed 52: starting validation-set evaluation...
```

```
Seed 52: validation-set evaluation complete.
```

```
Seed 52: starting testing-set evaluation...
```

```
Seed 52: testing-set evaluation complete.
```

```
SEED 52 - TRAINING RESULTS
```

```
Accuracy (%): 99.7080
```

```
Sensitivity (%): 99.5259
```

```
Specificity (%): 99.8914
```

```
Precision (%): 99.8918
```

```
F1-Score (%): 99.7085
```

```
Error Rate (%): 0.2920
```

```
ROC-AUC: 0.9995
```

```
SEED 52 - VALIDATION RESULTS
```

```
Accuracy (%): 97.3462
```

```
Sensitivity (%): 96.7480
```

```
Specificity (%): 98.2196
```

```
Precision (%): 98.7552
```

```
F1-Score (%): 97.7413
```

```
Error Rate (%): 2.6538
```

```
ROC-AUC: 0.9934
```

```
SEED 52 - TESTING RESULTS
=====
Accuracy (%): 96.0474
Sensitivity (%): 94.0299
Specificity (%): 98.3193
Precision (%): 98.4375
F1-Score (%): 96.1832
Error Rate (%): 3.9526
ROC-AUC: 0.9866

=====
SEED 52 - TRAINING CONFUSION MATRIX
=====
[[4601    5]
 [ 22 4618]]

=====
SEED 52 - VALIDATION CONFUSION MATRIX
=====
[[331    6]
 [ 16 476]]

=====
SEED 52 - TESTING CONFUSION MATRIX
=====
[[234    4]
 [ 16 252]]

=====
SEED 52 PERFORMANCE COMPARISON
=====
      Metric Training Validation Testing
Accuracy (%) 99.707982  97.346200 96.047431
Sensitivity (%) 99.525862 96.747967 94.029851
Specificity (%) 99.891446 98.219585 98.319328
Precision (%) 99.891845  98.755187 98.437500
F1-Score (%) 99.708518  97.741273 96.183206
Error Rate (%)  0.292018   2.653800  3.952569
ROC-AUC  0.999487   0.993432  0.986642

Seed 52 completion marker saved.
Seed 52 result saved: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/seed_results/seed_52_re

=====
FINISHED SEED 52
=====
Completed seeds: [42, 52]
Remaining seeds: [62, 72, 82]

#####
RUNNING SEED 62 (3/5)
#####

=====
STARTING SEED 62
=====

Random Seed: 62
Seed 62: creating DataLoaders...
Seed 62: DataLoaders ready | Train batches=145, Val batches=13, Test batches=8
Seed 62: creating EfficientNet-B0 model...
Seed 62: model created successfully.

No checkpoint found for Seed 62.
Starting Seed 62 from Epoch 1.

=====
SEED 62 | EPOCH 1/10 | STARTING TRAINING
=====

Seed 62 | Epoch 1/10: 100% 145/145 [01:05<00:00, 3.09batch/s, acc=87.04%, loss=0.5645]
Seed 62 | Epoch 1: FIRST BATCH COMPLETED in 0.45 seconds | Loss=0.6831

Seed 62 | Epoch 1: training finished. Starting validation...
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 62 | Epoch 1: validation finished.
```

```
-----
Seed 62 | Epoch 1/10
Train Loss: 0.5645 | Val Loss: 0.4964
Train Accuracy: 87.04% | Val Accuracy: 95.05%
Learning Rate: 0.00010000
Epoch Time: 1.27 minutes
Total Seed Time: 1.27 minutes
Best Val Loss: 0.4964 | Best Model: UPDATED
-----
```

```
Checkpoint saved successfully: Seed 62, Epoch 1
Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_62_checkpoint1.pth
```

```
=====
SEED 62 | EPOCH 2/10 | STARTING TRAINING
=====
```

```
Seed 62 | Epoch 2/10: 100% 145/145 [01:04<00:00, 2.35batch/s, acc=98.56%, loss=0.4701]
Seed 62 | Epoch 2: FIRST BATCH COMPLETED in 0.80 seconds | Loss=0.4943

Seed 62 | Epoch 2: training finished. Starting validation...
Seed 62 | Epoch 2: validation finished.
```

```
-----
Seed 62 | Epoch 2/10
Train Loss: 0.4701 | Val Loss: 0.4611
Train Accuracy: 98.56% | Val Accuracy: 95.05%
Learning Rate: 0.00010000
Epoch Time: 1.26 minutes
Total Seed Time: 2.53 minutes
Best Val Loss: 0.4611 | Best Model: UPDATED
-----
```

```
Checkpoint saved successfully: Seed 62, Epoch 2
Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_62_checkpoint2.pth
```

```
=====
SEED 62 | EPOCH 3/10 | STARTING TRAINING
=====
```

```
Seed 62 | Epoch 3/10: 100% 145/145 [01:06<00:00, 3.18batch/s, acc=99.17%, loss=0.4396]
Seed 62 | Epoch 3: FIRST BATCH COMPLETED in 0.71 seconds | Loss=0.4386

Seed 62 | Epoch 3: training finished. Starting validation...
Seed 62 | Epoch 3: validation finished.
```

```
-----
Seed 62 | Epoch 3/10
Train Loss: 0.4396 | Val Loss: 0.4323
Train Accuracy: 99.17% | Val Accuracy: 95.66%
Learning Rate: 0.00010000
Epoch Time: 1.29 minutes
Total Seed Time: 3.83 minutes
Best Val Loss: 0.4323 | Best Model: UPDATED
-----
```

```
Checkpoint saved successfully: Seed 62, Epoch 3
Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_62_checkpoint3.pth
```

```
=====
SEED 62 | EPOCH 4/10 | STARTING TRAINING
=====
```

```
Seed 62 | Epoch 4/10: 100% 145/145 [01:05<00:00, 2.51batch/s, acc=99.42%, loss=0.4249]
Seed 62 | Epoch 4: FIRST BATCH COMPLETED in 0.53 seconds | Loss=0.4286

Seed 62 | Epoch 4: training finished. Starting validation...
Seed 62 | Epoch 4: validation finished.
```

```
-----
Seed 62 | Epoch 4/10
Train Loss: 0.4249 | Val Loss: 0.4284
Train Accuracy: 99.42% | Val Accuracy: 95.90%
Learning Rate: 0.00010000
Epoch Time: 1.26 minutes
Total Seed Time: 5.10 minutes
Best Val Loss: 0.4284 | Best Model: UPDATED
-----
```

```
Checkpoint saved successfully: Seed 62, Epoch 4
Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_62_checkpoint4.pth
```

```
=====
SEED 62 | EPOCH 5/10 | STARTING TRAINING
=====
```

```
Seed 62 | Epoch 5/10: 100% 145/145 [01:05<00:00, 2.59batch/s, acc=99.49%, loss=0.4220]
Seed 62 | Epoch 5: FIRST BATCH COMPLETED in 0.44 seconds | Loss=0.4092

Seed 62 | Epoch 5: training finished. Starting validation...
Seed 62 | Epoch 5: validation finished.
```

```
-----
Seed 62 | Epoch 5/10
Train Loss: 0.4220 | Val Loss: 0.4247
Train Accuracy: 99.49% | Val Accuracy: 95.78%
Learning Rate: 0.00001000
Epoch Time: 1.25 minutes
Total Seed Time: 6.35 minutes
Best Val Loss: 0.4247 | Best Model: UPDATED
-----
```

```
Checkpoint saved successfully: Seed 62, Epoch 5
Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_62_checkpoint
```

```
=====
SEED 62 | EPOCH 6/10 | STARTING TRAINING
=====
```

```
Seed 62 | Epoch 6/10: 100% 145/145 [01:05<00:00, 3.07batch/s, acc=99.48%, loss=0.4196]
Seed 62 | Epoch 6: FIRST BATCH COMPLETED in 0.41 seconds | Loss=0.4479

Seed 62 | Epoch 6: training finished. Starting validation...
Seed 62 | Epoch 6: validation finished.
```

```
-----
Seed 62 | Epoch 6/10
Train Loss: 0.4196 | Val Loss: 0.4218
Train Accuracy: 99.48% | Val Accuracy: 96.26%
Learning Rate: 0.00001000
Epoch Time: 1.26 minutes
Total Seed Time: 7.62 minutes
Best Val Loss: 0.4218 | Best Model: UPDATED
-----
```

```
Checkpoint saved successfully: Seed 62, Epoch 6
Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_62_checkpoint
```

```
=====
SEED 62 | EPOCH 7/10 | STARTING TRAINING
=====
```

```
Seed 62 | Epoch 7/10: 100% 145/145 [01:05<00:00, 2.46batch/s, acc=99.50%, loss=0.4183]
Seed 62 | Epoch 7: FIRST BATCH COMPLETED in 0.41 seconds | Loss=0.4209

Seed 62 | Epoch 7: training finished. Starting validation...
Seed 62 | Epoch 7: validation finished.
```

```
-----
Seed 62 | Epoch 7/10
Train Loss: 0.4183 | Val Loss: 0.4213
Train Accuracy: 99.50% | Val Accuracy: 96.26%
Learning Rate: 0.00001000
Epoch Time: 1.25 minutes
Total Seed Time: 8.88 minutes
Best Val Loss: 0.4213 | Best Model: UPDATED
-----
```

```
Checkpoint saved successfully: Seed 62, Epoch 7
Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_62_checkpoint
```

```
=====
SEED 62 | EPOCH 8/10 | STARTING TRAINING
=====
```

```
Seed 62 | Epoch 8/10: 100% 145/145 [01:05<00:00, 2.98batch/s, acc=99.47%, loss=0.4177]
Seed 62 | Epoch 8: FIRST BATCH COMPLETED in 0.49 seconds | Loss=0.4265

Seed 62 | Epoch 8: training finished. Starting validation...
Seed 62 | Epoch 8: validation finished.
```

```
-----
Seed 62 | Epoch 8/10
Train Loss: 0.4177 | Val Loss: 0.4207
Train Accuracy: 99.47% | Val Accuracy: 96.14%
```

Learning Rate: 0.0000100
 Epoch Time: 1.28 minutes
 Total Seed Time: 10.17 minutes
 Best Val Loss: 0.4207 | Best Model: UPDATED

Checkpoint saved successfully: Seed 62, Epoch 8
 Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_62_checkpoint

=====

SEED 62 | EPOCH 9/10 | STARTING TRAINING

=====

Seed 62 | Epoch 9/10: 100% 145/145 [01:04<00:00, 3.36batch/s, acc=99.49%, loss=0.4179]

Seed 62 | Epoch 9: FIRST BATCH COMPLETED in 0.51 seconds | Loss=0.4439

Seed 62 | Epoch 9: training finished. Starting validation...
 Seed 62 | Epoch 9: validation finished.

Seed 62 | Epoch 9/10
 Train Loss: 0.4179 | Val Loss: 0.4205
 Train Accuracy: 99.49% | Val Accuracy: 96.14%
 Learning Rate: 0.0000100
 Epoch Time: 1.26 minutes
 Total Seed Time: 11.43 minutes
 Best Val Loss: 0.4205 | Best Model: UPDATED

Checkpoint saved successfully: Seed 62, Epoch 9
 Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_62_checkpoint

=====

SEED 62 | EPOCH 10/10 | STARTING TRAINING

=====

Seed 62 | Epoch 10/10: 100% 145/145 [01:05<00:00, 3.17batch/s, acc=99.56%, loss=0.4174]

Seed 62 | Epoch 10: FIRST BATCH COMPLETED in 0.46 seconds | Loss=0.4059

Seed 62 | Epoch 10: training finished. Starting validation...
 Seed 62 | Epoch 10: validation finished.

Seed 62 | Epoch 10/10
 Train Loss: 0.4174 | Val Loss: 0.4208
 Train Accuracy: 99.56% | Val Accuracy: 96.14%
 Learning Rate: 0.0000100
 Epoch Time: 1.25 minutes
 Total Seed Time: 12.68 minutes
 Best Val Loss: 0.4205 | Best Model: NOT UPDATED

Checkpoint saved successfully: Seed 62, Epoch 10
 Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_62_checkpoint

=====

SEED 62 TRAINING COMPLETE

=====

Total time: 12.69 minutes
 Best validation-loss model loaded for Seed 62.

Seed 62: starting final training-set evaluation...
 Seed 62: training-set evaluation complete.
 Seed 62: starting validation-set evaluation...
 Seed 62: validation-set evaluation complete.
 Seed 62: starting testing-set evaluation...
 Seed 62: testing-set evaluation complete.

=====

SEED 62 - TRAINING RESULTS

=====

Accuracy (%): 99.5782
 Sensitivity (%): 99.3966
 Specificity (%): 99.7612
 Precision (%): 99.7621
 F1-Score (%): 99.5790
 Error Rate (%): 0.4218
 ROC-AUC: 0.9997

=====

SEED 62 - VALIDATION RESULTS

=====

Accuracy (%): 96.1399
 Sensitivity (%): 95.1220
 Specificity (%): 97.6261
 Precision (%): 98.3193
 F1-Score (%): 96.6942
 Error Rate (%): 3.8601
 ROC-AUC: 0.9933

=====

SEED 62 - TESTING RESULTS

=====

Accuracy (%): 94.4664
 Sensitivity (%): 91.0448
 Specificity (%): 98.3193
 Precision (%): 98.3871
 F1-Score (%): 94.5736
 Error Rate (%): 5.5336
 ROC-AUC: 0.9916

=====

SEED 62 - TRAINING CONFUSION MATRIX

=====

```
[[4595  11]
 [ 28 4612]]
```

=====

SEED 62 - VALIDATION CONFUSION MATRIX

=====

```
[[329   8]
 [ 24 468]]
```

=====

SEED 62 - TESTING CONFUSION MATRIX

=====

```
[[234   4]
 [ 24 244]]
```

=====

SEED 62 PERFORMANCE COMPARISON

=====

Metric	Training	Validation	Testing
Accuracy (%)	99.578196	96.139928	94.466403
Sensitivity (%)	99.396552	95.121951	91.044776
Specificity (%)	99.761181	97.626113	98.319328
Precision (%)	99.762059	98.319328	98.387097
F1-Score (%)	99.578970	96.694215	94.573643
Error Rate (%)	0.421804	3.860072	5.533597
ROC-AUC	0.999652	0.993251	0.991628

Seed 62 completion marker saved.

Seed 62 result saved: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/seed_results/seed_62_re

=====

FINISHED SEED 62

=====

Completed seeds: [42, 52, 62]
 Remaining seeds: [72, 82]

#####

RUNNING SEED 72 (4/5)

#####

=====

STARTING SEED 72

=====

Random Seed: 72
 Seed 72: creating DataLoaders...
 Seed 72: DataLoaders ready | Train batches=145, Val batches=13, Test batches=8
 Seed 72: creating EfficientNet-B0 model...
 Seed 72: model created successfully.

No checkpoint found for Seed 72.
 Starting Seed 72 from Epoch 1.

SEED 72 | EPOCH 1/10 | STARTING TRAINING

Seed 72 | Epoch 1/10: 100%

145/145 [01:02<00:00, 3.02batch/s, acc=49.82%, loss=0.6204]

Seed 72 | Epoch 1: FIRST BATCH COMPLETED in 0.57 seconds | Loss=0.7080

Seed 72 | Epoch 1: training finished. Starting validation...

/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA images
warnings.warn(

Seed 72 | Epoch 1: validation finished.

Seed 72 | Epoch 1/10

Train Loss: 0.6204 | Val Loss: 0.6056

Train Accuracy: 49.82% | Val Accuracy: 40.65%

Learning Rate: 0.00010000

Epoch Time: 1.20 minutes

Total Seed Time: 1.21 minutes

Best Val Loss: 0.6056 | Best Model: UPDATED

Checkpoint saved successfully: Seed 72, Epoch 1

Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_72_checkpoint1.pth

SEED 72 | EPOCH 2/10 | STARTING TRAINING

Seed 72 | Epoch 2/10: 100%

145/145 [01:01<00:00, 2.98batch/s, acc=49.82%, loss=0.5256]

Seed 72 | Epoch 2: FIRST BATCH COMPLETED in 0.43 seconds | Loss=0.5322

Seed 72 | Epoch 2: training finished. Starting validation...

Seed 72 | Epoch 2: validation finished.

Seed 72 | Epoch 2/10

Train Loss: 0.5256 | Val Loss: 0.5583

Train Accuracy: 49.82% | Val Accuracy: 40.65%

Learning Rate: 0.00010000

Epoch Time: 1.21 minutes

Total Seed Time: 2.43 minutes

Best Val Loss: 0.5583 | Best Model: UPDATED

Checkpoint saved successfully: Seed 72, Epoch 2

Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_72_checkpoint2.pth

SEED 72 | EPOCH 3/10 | STARTING TRAINING

Seed 72 | Epoch 3/10: 100%

145/145 [01:07<00:00, 2.54batch/s, acc=82.09%, loss=0.4872]

Seed 72 | Epoch 3: FIRST BATCH COMPLETED in 0.50 seconds | Loss=0.4998

Seed 72 | Epoch 3: training finished. Starting validation...

Seed 72 | Epoch 3: validation finished.

Seed 72 | Epoch 3/10

Train Loss: 0.4872 | Val Loss: 0.5160

Train Accuracy: 82.09% | Val Accuracy: 97.59%

Learning Rate: 0.00010000

Epoch Time: 1.32 minutes

Total Seed Time: 3.75 minutes

Best Val Loss: 0.5160 | Best Model: UPDATED

Checkpoint saved successfully: Seed 72, Epoch 3

Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_72_checkpoint3.pth

SEED 72 | EPOCH 4/10 | STARTING TRAINING

Seed 72 | Epoch 4/10: 100%

145/145 [01:07<00:00, 2.97batch/s, acc=98.52%, loss=0.4679]

Seed 72 | Epoch 4: FIRST BATCH COMPLETED in 0.84 seconds | Loss=0.4739

Seed 72 | Epoch 4: training finished. Starting validation...

Seed 72 | Epoch 4: validation finished.

Seed 72 | Epoch 4/10

Train Loss: 0.4679 | Val Loss: 0.5111

Train Accuracy: 98.52% | Val Accuracy: 97.83%
Learning Rate: 0.00001000
Epoch Time: 1.32 minutes
Total Seed Time: 5.08 minutes
Best Val Loss: 0.5111 | Best Model: UPDATED

Checkpoint saved successfully: Seed 72, Epoch 4

Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_72_checkpoint

=====

SEED 72 | EPOCH 5/10 | STARTING TRAINING

=====

Seed 72 | Epoch 5/10: 99%

143/145 [01:11<00:00, 2.12batch/s, acc=98.80%, loss=0.4633]

Seed 72 | Epoch 5: FIRST BATCH COMPLETED in 0.42 seconds | Loss=0.4351

Seed 72 | Epoch 5: training finished. Starting validation...

Seed 72 | Epoch 5: validation finished.

Seed 72 | Epoch 5/10
Train Loss: 0.4633 | Val Loss: 0.5073
Train Accuracy: 98.81% | Val Accuracy: 97.59%
Learning Rate: 0.00001000
Epoch Time: 1.40 minutes
Total Seed Time: 6.48 minutes
Best Val Loss: 0.5073 | Best Model: UPDATED

Checkpoint saved successfully: Seed 72, Epoch 5

Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_72_checkpoint

=====

SEED 72 | EPOCH 6/10 | STARTING TRAINING

=====

Seed 72 | Epoch 6/10: 100%

145/145 [01:13<00:00, 2.64batch/s, acc=98.85%, loss=0.4593]

Seed 72 | Epoch 6: FIRST BATCH COMPLETED in 0.58 seconds | Loss=0.4939

Seed 72 | Epoch 6: training finished. Starting validation...

Seed 72 | Epoch 6: validation finished.

Seed 72 | Epoch 6/10
Train Loss: 0.4593 | Val Loss: 0.5036
Train Accuracy: 98.85% | Val Accuracy: 97.95%
Learning Rate: 0.00001000
Epoch Time: 1.40 minutes
Total Seed Time: 7.88 minutes
Best Val Loss: 0.5036 | Best Model: UPDATED

Checkpoint saved successfully: Seed 72, Epoch 6

Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_72_checkpoint

=====

SEED 72 | EPOCH 7/10 | STARTING TRAINING

=====

Seed 72 | Epoch 7/10: 100%

145/145 [01:05<00:00, 2.45batch/s, acc=98.99%, loss=0.4579]

Seed 72 | Epoch 7: FIRST BATCH COMPLETED in 0.47 seconds | Loss=0.4492

Seed 72 | Epoch 7: training finished. Starting validation...

Seed 72 | Epoch 7: validation finished.

Seed 72 | Epoch 7/10
Train Loss: 0.4579 | Val Loss: 0.5032
Train Accuracy: 98.99% | Val Accuracy: 97.95%
Learning Rate: 0.00001000
Epoch Time: 1.26 minutes
Total Seed Time: 9.15 minutes
Best Val Loss: 0.5032 | Best Model: UPDATED

Checkpoint saved successfully: Seed 72, Epoch 7

Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_72_checkpoint

=====

SEED 72 | EPOCH 8/10 | STARTING TRAINING

=====

Seed 72 | Epoch 8/10: 100%

145/145 [01:06<00:00, 3.06batch/s, acc=98.92%, loss=0.4584]

Seed 72 | Epoch 8: FIRST BATCH COMPLETED in 0.41 seconds | Loss=0.4694

Seed 72 | Epoch 8: training finished. Starting validation...

Seed 72 | Epoch 8: validation finished.

```
-----
Seed 72 | Epoch 8/10
Train Loss: 0.4584 | Val Loss: 0.5021
Train Accuracy: 98.92% | Val Accuracy: 97.95%
Learning Rate: 0.00000100
Epoch Time: 1.28 minutes
Total Seed Time: 10.44 minutes
Best Val Loss: 0.5021 | Best Model: UPDATED
-----
```

Checkpoint saved successfully: Seed 72, Epoch 8

Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_72_checkpoint

```
=====
SEED 72 | EPOCH 9/10 | STARTING TRAINING
=====
```

Seed 72 | Epoch 9/10: 100%

145/145 [01:03<00:00, 2.93batch/s, acc=99.12%, loss=0.4569]

Seed 72 | Epoch 9: FIRST BATCH COMPLETED in 0.40 seconds | Loss=0.4404

Seed 72 | Epoch 9: training finished. Starting validation...

Seed 72 | Epoch 9: validation finished.

```
-----
Seed 72 | Epoch 9/10
Train Loss: 0.4569 | Val Loss: 0.5027
Train Accuracy: 99.12% | Val Accuracy: 97.95%
Learning Rate: 0.00000010
Epoch Time: 1.23 minutes
Total Seed Time: 11.68 minutes
Best Val Loss: 0.5021 | Best Model: NOT UPDATED
-----
```

Checkpoint saved successfully: Seed 72, Epoch 9

Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_72_checkpoint

```
=====
SEED 72 | EPOCH 10/10 | STARTING TRAINING
=====
```

Seed 72 | Epoch 10/10: 100%

145/145 [01:04<00:00, 1.86batch/s, acc=98.98%, loss=0.4575]

Seed 72 | Epoch 10: FIRST BATCH COMPLETED in 0.41 seconds | Loss=0.4472

Seed 72 | Epoch 10: training finished. Starting validation...

Seed 72 | Epoch 10: validation finished.

```
-----
Seed 72 | Epoch 10/10
Train Loss: 0.4575 | Val Loss: 0.5022
Train Accuracy: 98.98% | Val Accuracy: 97.83%
Learning Rate: 0.00000010
Epoch Time: 1.23 minutes
Total Seed Time: 12.92 minutes
Best Val Loss: 0.5021 | Best Model: NOT UPDATED
-----
```

Checkpoint saved successfully: Seed 72, Epoch 10

Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_72_checkpoint

```
=====
SEED 72 TRAINING COMPLETE
=====
```

Total time: 12.92 minutes

Best validation-loss model loaded for Seed 72.

Seed 72: starting final training-set evaluation...

Seed 72: training-set evaluation complete.

Seed 72: starting validation-set evaluation...

Seed 72: validation-set evaluation complete.

Seed 72: starting testing-set evaluation...

Seed 72: testing-set evaluation complete.

```
=====
SEED 72 - TRAINING RESULTS
=====
```

Accuracy (%): 99.1564

Sensitivity (%): 98.5560

```
07/09/2026, 18:38                                output EFFICIENTNET-B0 + NONLINEAR CLASSICAL MLP.ipynb - Colab

Specificity (%): 99.7612
Precision (%): 99.7600
F1-Score (%): 99.1544
Error Rate (%): 0.8436
ROC-AUC: 0.9998

=====
SEED 72 - VALIDATION RESULTS
=====
Accuracy (%): 97.9493
Sensitivity (%): 97.9675
Specificity (%): 97.9228
Precision (%): 98.5685
F1-Score (%): 98.2671
Error Rate (%): 2.0507
ROC-AUC: 0.9935

=====
SEED 72 - TESTING RESULTS
=====
Accuracy (%): 97.2332
Sensitivity (%): 96.2687
Specificity (%): 98.3193
Precision (%): 98.4733
F1-Score (%): 97.3585
Error Rate (%): 2.7668
ROC-AUC: 0.9962

=====
SEED 72 - TRAINING CONFUSION MATRIX
=====
[[4595  11]
 [ 67 4573]]

=====
SEED 72 - VALIDATION CONFUSION MATRIX
=====
[[330  7]
 [ 10 482]]

=====
SEED 72 - TESTING CONFUSION MATRIX
=====
[[234  4]
 [ 10 258]]

=====
SEED 72 PERFORMANCE COMPARISON
=====
Metric Training Validation Testing
Accuracy (%) 99.156392 97.949337 97.233202
Sensitivity (%) 98.556034 97.967480 96.268657
Specificity (%) 99.761181 97.922849 98.319328
Precision (%) 99.760035 98.568507 98.473282
F1-Score (%) 99.154380 98.267074 97.358491
Error Rate (%) 0.843608 2.050663 2.766798
ROC-AUC 0.999820 0.993529 0.996206

Seed 72 completion marker saved.
Seed 72 result saved: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/seed_results/seed_72_re

=====
FINISHED SEED 72
=====
Completed seeds: [42, 52, 62, 72]
Remaining seeds: [82]

#####
RUNNING SEED 82 (5/5)
#####

=====
STARTING SEED 82
=====

Random Seed: 82
Seed 82: creating DataLoaders...
```

Seed 82: DataLoaders ready | Train batches=145, Val batches=13, Test batches=8
Seed 82: creating EfficientNet-B0 model...
Seed 82: model created successfully.

No checkpoint found for Seed 82.
Starting Seed 82 from Epoch 1.

=====

SEED 82 | EPOCH 1/10 | STARTING TRAINING

=====

Seed 82 | Epoch 1/10: 100% 145/145 [01:02<00:00, 2.59batch/s, acc=50.24%, loss=0.5688]

Seed 82 | Epoch 1: FIRST BATCH COMPLETED in 0.51 seconds | Loss=0.6771

Seed 82 | Epoch 1: training finished. Starting validation...
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA images
warnings.warn(
Seed 82 | Epoch 1: validation finished.

Seed 82 | Epoch 1/10
Train Loss: 0.5688 | Val Loss: 0.4560
Train Accuracy: 50.24% | Val Accuracy: 67.79%
Learning Rate: 0.00010000
Epoch Time: 1.23 minutes
Total Seed Time: 1.23 minutes
Best Val Loss: 0.4560 | Best Model: UPDATED

Checkpoint saved successfully: Seed 82, Epoch 1
Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_82_checkpoint

=====

SEED 82 | EPOCH 2/10 | STARTING TRAINING

=====

Seed 82 | Epoch 2/10: 100% 145/145 [01:02<00:00, 2.93batch/s, acc=94.52%, loss=0.4554]

Seed 82 | Epoch 2: FIRST BATCH COMPLETED in 0.62 seconds | Loss=0.5168

Seed 82 | Epoch 2: training finished. Starting validation...
Seed 82 | Epoch 2: validation finished.

Seed 82 | Epoch 2/10
Train Loss: 0.4554 | Val Loss: 0.4161
Train Accuracy: 94.52% | Val Accuracy: 95.78%
Learning Rate: 0.00010000
Epoch Time: 1.22 minutes
Total Seed Time: 2.45 minutes
Best Val Loss: 0.4161 | Best Model: UPDATED

Checkpoint saved successfully: Seed 82, Epoch 2
Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_82_checkpoint

=====

SEED 82 | EPOCH 3/10 | STARTING TRAINING

=====

Seed 82 | Epoch 3/10: 100% 145/145 [01:04<00:00, 2.29batch/s, acc=99.03%, loss=0.4211]

Seed 82 | Epoch 3: FIRST BATCH COMPLETED in 0.57 seconds | Loss=0.4752

Seed 82 | Epoch 3: training finished. Starting validation...
Seed 82 | Epoch 3: validation finished.

Seed 82 | Epoch 3/10
Train Loss: 0.4211 | Val Loss: 0.3829
Train Accuracy: 99.03% | Val Accuracy: 96.62%
Learning Rate: 0.00001000
Epoch Time: 1.24 minutes
Total Seed Time: 3.70 minutes
Best Val Loss: 0.3829 | Best Model: UPDATED

Checkpoint saved successfully: Seed 82, Epoch 3
Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_82_checkpoint

=====

SEED 82 | EPOCH 4/10 | STARTING TRAINING

=====

Seed 82 | Epoch 4/10: 100% 145/145 [01:05<00:00, 2.91batch/s, acc=99.04%, loss=0.4057]

Seed 82 | Epoch 4: FIRST BATCH COMPLETED in 0.43 seconds | Loss=0.4399

Seed 82 | Epoch 4: training finished. Starting validation...

Seed 82 | Epoch 4: validation finished.

Seed 82 | Epoch 4/10
Train Loss: 0.4057 | Val Loss: 0.3759
Train Accuracy: 99.04% | Val Accuracy: 97.23%
Learning Rate: 0.00001000
Epoch Time: 1.27 minutes
Total Seed Time: 4.97 minutes
Best Val Loss: 0.3759 | Best Model: UPDATED

Checkpoint saved successfully: Seed 82, Epoch 4

Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_82_checkpoint

=====

SEED 82 | EPOCH 5/10 | STARTING TRAINING

=====

Seed 82 | Epoch 5/10: 100%

145/145 [01:03<00:00, 3.09batch/s, acc=99.49%, loss=0.4003]

Seed 82 | Epoch 5: FIRST BATCH COMPLETED in 0.52 seconds | Loss=0.3949

Seed 82 | Epoch 5: training finished. Starting validation...

Seed 82 | Epoch 5: validation finished.

Seed 82 | Epoch 5/10
Train Loss: 0.4003 | Val Loss: 0.3721
Train Accuracy: 99.49% | Val Accuracy: 97.71%
Learning Rate: 0.00001000
Epoch Time: 1.24 minutes
Total Seed Time: 6.21 minutes
Best Val Loss: 0.3721 | Best Model: UPDATED

Checkpoint saved successfully: Seed 82, Epoch 5

Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_82_checkpoint

=====

SEED 82 | EPOCH 6/10 | STARTING TRAINING

=====

Seed 82 | Epoch 6/10: 100%

145/145 [01:04<00:00, 2.47batch/s, acc=99.48%, loss=0.3979]

Seed 82 | Epoch 6: FIRST BATCH COMPLETED in 0.52 seconds | Loss=0.3828

Seed 82 | Epoch 6: training finished. Starting validation...

Seed 82 | Epoch 6: validation finished.

Seed 82 | Epoch 6/10
Train Loss: 0.3979 | Val Loss: 0.3685
Train Accuracy: 99.48% | Val Accuracy: 97.71%
Learning Rate: 0.00001000
Epoch Time: 1.24 minutes
Total Seed Time: 7.46 minutes
Best Val Loss: 0.3685 | Best Model: UPDATED

Checkpoint saved successfully: Seed 82, Epoch 6

Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_82_checkpoint

=====

SEED 82 | EPOCH 7/10 | STARTING TRAINING

=====

Seed 82 | Epoch 7/10: 100%

145/145 [01:05<00:00, 3.02batch/s, acc=99.39%, loss=0.3971]

Seed 82 | Epoch 7: FIRST BATCH COMPLETED in 0.42 seconds | Loss=0.3415

Seed 82 | Epoch 7: training finished. Starting validation...

Seed 82 | Epoch 7: validation finished.

Seed 82 | Epoch 7/10
Train Loss: 0.3971 | Val Loss: 0.3678
Train Accuracy: 99.39% | Val Accuracy: 97.95%
Learning Rate: 0.00001000
Epoch Time: 1.27 minutes
Total Seed Time: 8.73 minutes
Best Val Loss: 0.3678 | Best Model: UPDATED

```
-----
Checkpoint saved successfully: Seed 82, Epoch 7
Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_82_checkpoint7.pth

=====
SEED 82 | EPOCH 8/10 | STARTING TRAINING
=====

Seed 82 | Epoch 8/10: 100% 145/145 [01:04<00:00, 3.33batch/s, acc=99.49%, loss=0.3968]
Seed 82 | Epoch 8: FIRST BATCH COMPLETED in 0.52 seconds | Loss=0.4223

Seed 82 | Epoch 8: training finished. Starting validation...
Seed 82 | Epoch 8: validation finished.

-----
Seed 82 | Epoch 8/10
Train Loss: 0.3968 | Val Loss: 0.3691
Train Accuracy: 99.49% | Val Accuracy: 98.07%
Learning Rate: 0.00000100
Epoch Time: 1.25 minutes
Total Seed Time: 9.99 minutes
Best Val Loss: 0.3678 | Best Model: NOT UPDATED

-----
Checkpoint saved successfully: Seed 82, Epoch 8
Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_82_checkpoint8.pth

=====
SEED 82 | EPOCH 9/10 | STARTING TRAINING
=====

Seed 82 | Epoch 9/10: 100% 145/145 [01:03<00:00, 2.70batch/s, acc=99.61%, loss=0.3957]
Seed 82 | Epoch 9: FIRST BATCH COMPLETED in 0.44 seconds | Loss=0.3862

Seed 82 | Epoch 9: training finished. Starting validation...
Seed 82 | Epoch 9: validation finished.

-----
Seed 82 | Epoch 9/10
Train Loss: 0.3957 | Val Loss: 0.3682
Train Accuracy: 99.61% | Val Accuracy: 98.07%
Learning Rate: 0.00000010
Epoch Time: 1.22 minutes
Total Seed Time: 11.22 minutes
Best Val Loss: 0.3678 | Best Model: NOT UPDATED

-----
Checkpoint saved successfully: Seed 82, Epoch 9
Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_82_checkpoint9.pth

=====
SEED 82 | EPOCH 10/10 | STARTING TRAINING
=====

Seed 82 | Epoch 10/10: 100% 145/145 [01:04<00:00, 3.08batch/s, acc=99.49%, loss=0.3966]
Seed 82 | Epoch 10: FIRST BATCH COMPLETED in 0.52 seconds | Loss=0.4017

Seed 82 | Epoch 10: training finished. Starting validation...
Seed 82 | Epoch 10: validation finished.

-----
Seed 82 | Epoch 10/10
Train Loss: 0.3966 | Val Loss: 0.3670
Train Accuracy: 99.49% | Val Accuracy: 97.71%
Learning Rate: 0.00000010
Epoch Time: 1.25 minutes
Total Seed Time: 12.48 minutes
Best Val Loss: 0.3670 | Best Model: UPDATED

-----
Checkpoint saved successfully: Seed 82, Epoch 10
Checkpoint path: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/checkpoints/seed_82_checkpoint10.pth

=====
SEED 82 TRAINING COMPLETE
=====
Total time: 12.48 minutes
Best validation-loss model loaded for Seed 82.

Seed 82: starting final training-set evaluation...
Seed 82: training-set evaluation complete.
Seed 82: starting validation-set evaluation...
```



```
07/09/2026, 18:38                                output EFFICIENTNET-B0 + NONLINEAR CLASSICAL MLP.ipynb - Colab

Seed 82: validation-set evaluation complete.
Seed 82: starting testing-set evaluation...
Seed 82: testing-set evaluation complete.

=====
SEED 82 - TRAINING RESULTS
=====
Accuracy (%): 99.7729
Sensitivity (%): 99.7414
Specificity (%): 99.8046
Precision (%): 99.8059
F1-Score (%): 99.7736
Error Rate (%): 0.2271
ROC-AUC: 0.9998

=====
SEED 82 - VALIDATION RESULTS
=====
Accuracy (%): 97.7081
Sensitivity (%): 98.3740
Specificity (%): 96.7359
Precision (%): 97.7778
F1-Score (%): 98.0750
Error Rate (%): 2.2919
ROC-AUC: 0.9960

=====
SEED 82 - TESTING RESULTS
=====
Accuracy (%): 96.6403
Sensitivity (%): 97.0149
Specificity (%): 96.2185
Precision (%): 96.6543
F1-Score (%): 96.8343
Error Rate (%): 3.3597
ROC-AUC: 0.9938

=====
SEED 82 - TRAINING CONFUSION MATRIX
=====
[[4597   9]
 [ 12 4628]]

=====
SEED 82 - VALIDATION CONFUSION MATRIX
=====
[[326  11]
 [   8 484]]

=====
SEED 82 - TESTING CONFUSION MATRIX
=====
[[229   9]
 [   8 260]]

=====
SEED 82 PERFORMANCE COMPARISON
=====
Metric Training Validation Testing
Accuracy (%) 99.772875  97.708082 96.640316
Sensitivity (%) 99.741379 98.373984 97.014925
Specificity (%) 99.804603 96.735905 96.218487
Precision (%) 99.805909 97.777778 96.654275
F1-Score (%) 99.773634 98.074975 96.834264
Error Rate (%) 0.227125  2.291918  3.359684
ROC-AUC 0.999770  0.995971  0.993823

Seed 82 completion marker saved.
Seed 82 result saved: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/seed_results/seed_82_r

=====
FINISHED SEED 82
=====
Completed seeds: [42, 52, 62, 72, 82]
Remaining seeds: []

=====
```

```
FIVE-SEED EXPERIMENT STATUS
=====
Completed seeds: [42, 52, 62, 72, 82]
All five seeds completed successfully.

=====
INDIVIDUAL FIVE-SEED RESULTS
=====
Seed   Dataset  Accuracy (%)  Sensitivity (%)  Specificity (%)  Precision (%)  F1-Score (%)  Error Rate (%)  ROC-AUC
42     Training  99.686351     99.568966       99.804603       99.805574     99.687129     0.313649     0.999610
42     Validation 97.346200     96.747967       98.219585       98.755187     97.741273     2.653800     0.994741
42     Testing   96.047431     94.029851       98.319328       98.437500     96.183206     3.952569     0.992788
52     Training  99.707982     99.525862       99.891446       99.891845     99.708518     0.292018     0.999487
52     Validation 97.346200     96.747967       98.219585       98.755187     97.741273     2.653800     0.993432
52     Testing   96.047431     94.029851       98.319328       98.437500     96.183206     3.952569     0.986642
62     Training  99.578196     99.396552       99.761181       99.762059     99.578970     0.421804     0.999652
62     Validation 96.139928     95.121951       97.626113       98.319328     96.694215     3.860072     0.993251
62     Testing   94.466403     91.044776       98.319328       98.387097     94.573643     5.533597     0.991628
72     Training  99.156392     98.556034       99.761181       99.760035     99.154380     0.843608     0.999820
72     Validation 97.949337     97.967480       97.922849       98.568507     98.267074     2.050663     0.993529
72     Testing   97.233202     96.268657       98.319328       98.473282     97.358491     2.766798     0.996206
82     Training  99.772875     99.741379       99.804603       99.805909     99.773634     0.227125     0.999770
82     Validation 97.708082     98.373984       96.735905       97.777778     98.074975     2.291918     0.995971
82     Testing   96.640316     97.014925       96.218487       96.654275     96.834264     3.359684     0.993823

=====
PERFORMANCE COMPARISON TABLE - MEAN ± SD
=====
Dataset Accuracy (%) Mean ± SD Sensitivity (%) Mean ± SD Specificity (%) Mean ± SD Precision (%) Mean ± SD F1-Score (%) Mean ± SD Error Rate
Training  99.5804 ± 0.2472     99.3578 ± 0.4648     99.8046 ± 0.0532     99.8051 ± 0.0534     99.5805 ± 0.2483     0
Validation 97.2979 ± 0.6961     96.9919 ± 1.2726     97.7448 ± 0.6153     98.4352 ± 0.4087     97.7038 ± 0.6077     2
Testing   96.0870 ± 1.0307     94.4776 ± 2.3362     97.8992 ± 0.9395     98.0779 ± 0.7964     96.2266 ± 1.0472     3

=====
MEAN CONFUSION MATRIX - TRAINING
=====
[[4597.    9. ]
 [ 29.8 4610.2]]

Training CM SD:
[[ 2.45  2.45]
 [21.57 21.57]]

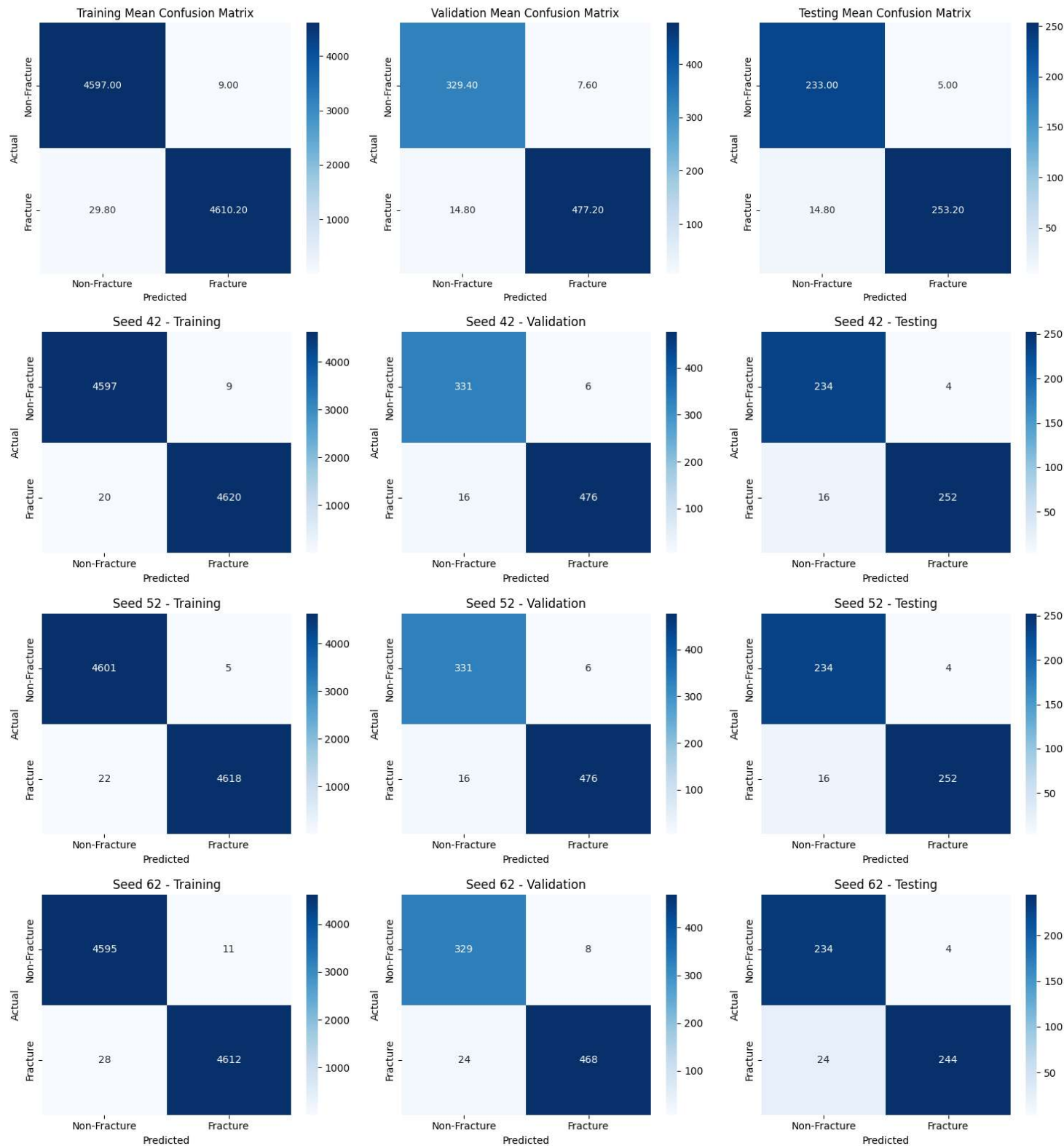
=====
MEAN CONFUSION MATRIX - VALIDATION
=====
[[329.4    7.6]
 [ 14.8 477.2]]

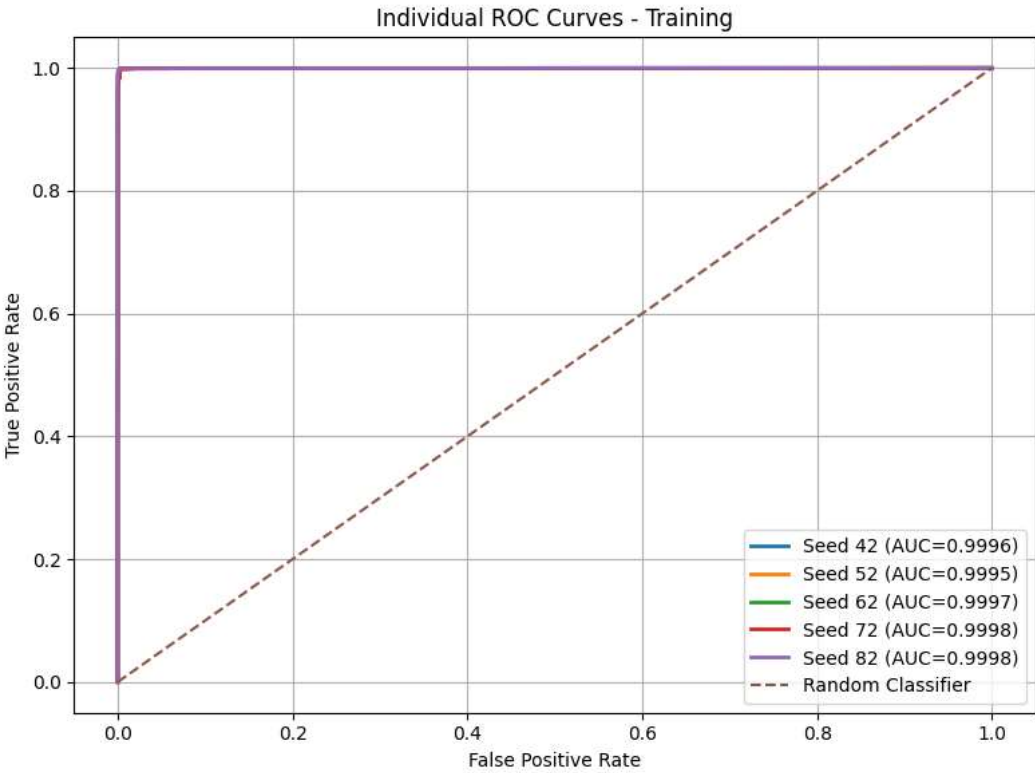
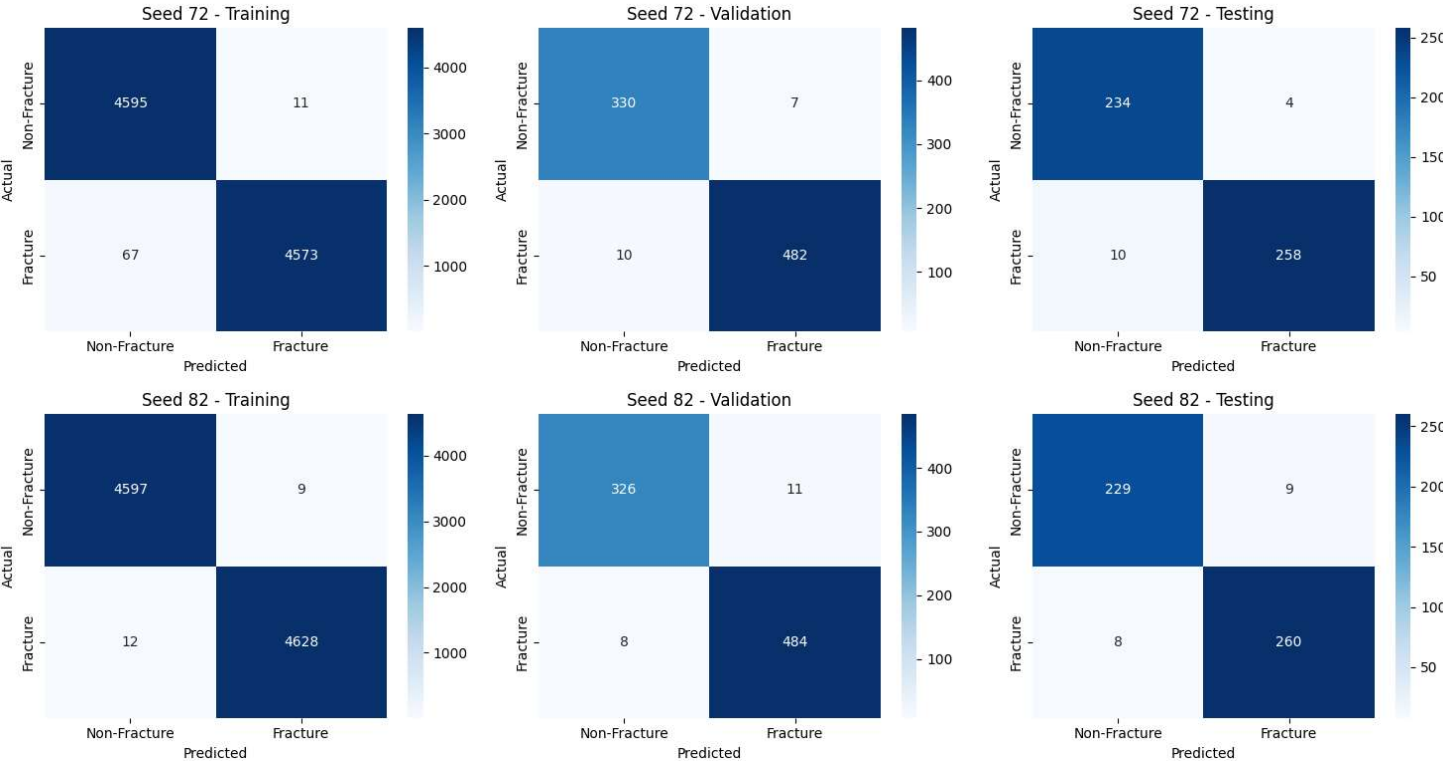
Validation CM SD:
[[2.07 2.07]
 [6.26 6.26]]

=====
MEAN CONFUSION MATRIX - TESTING
=====
[[233.    5. ]
 [ 14.8 253.2]]

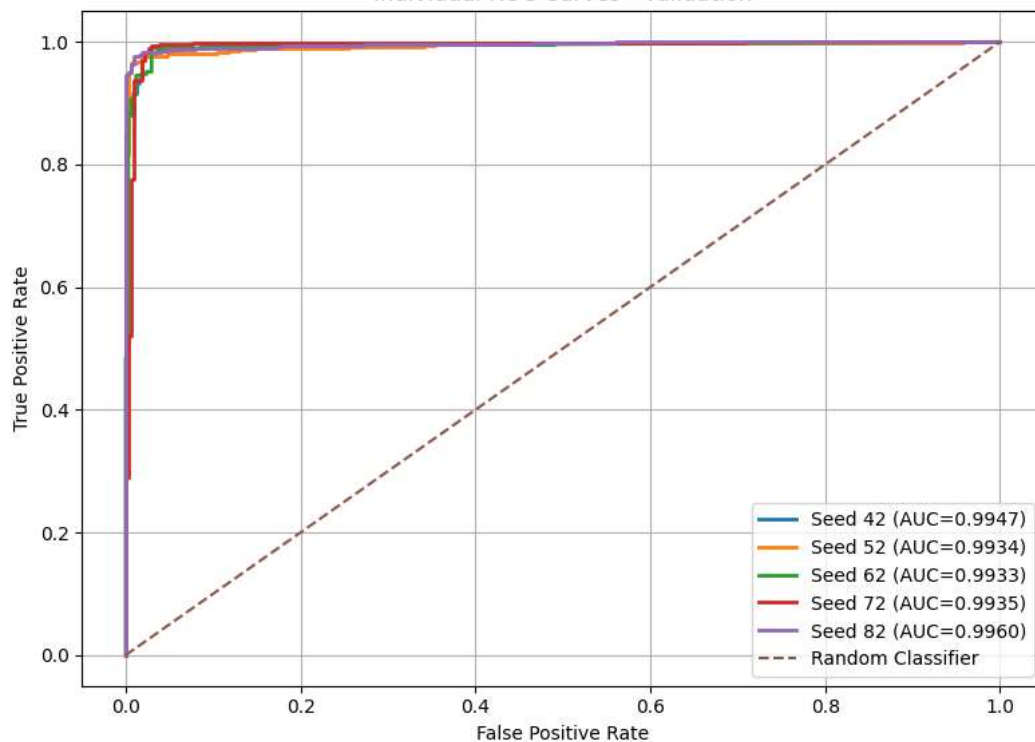
Testing CM SD:
[[2.24 2.24]
 [6.26 6.26]]
```

Mean Confusion Matrices Across Five Seeds

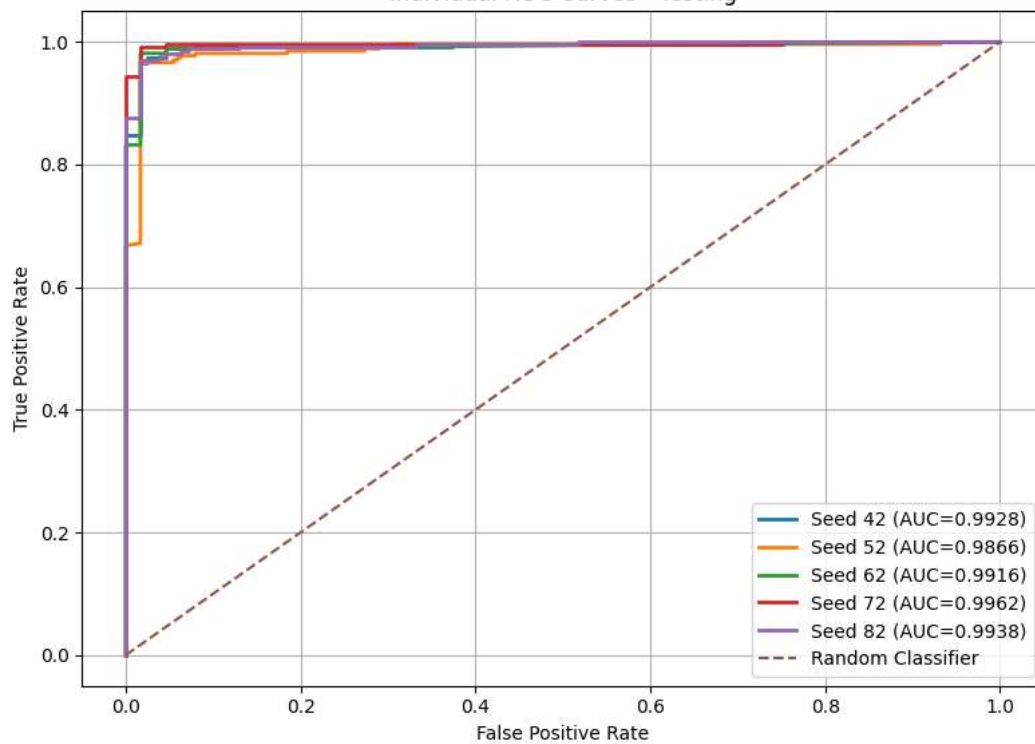


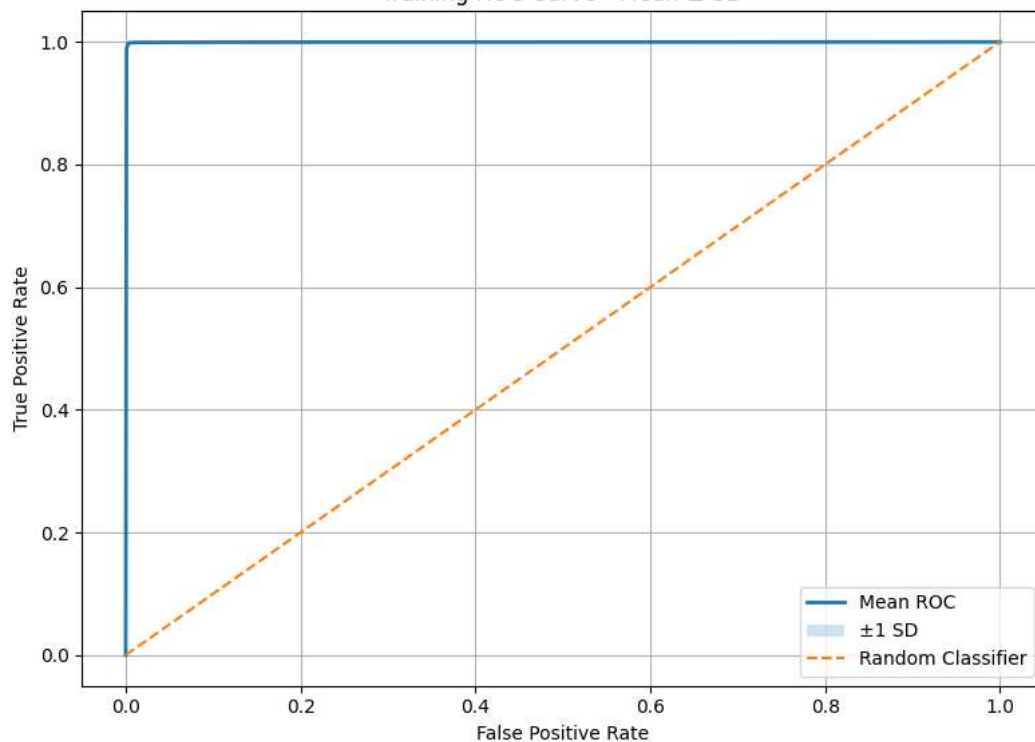
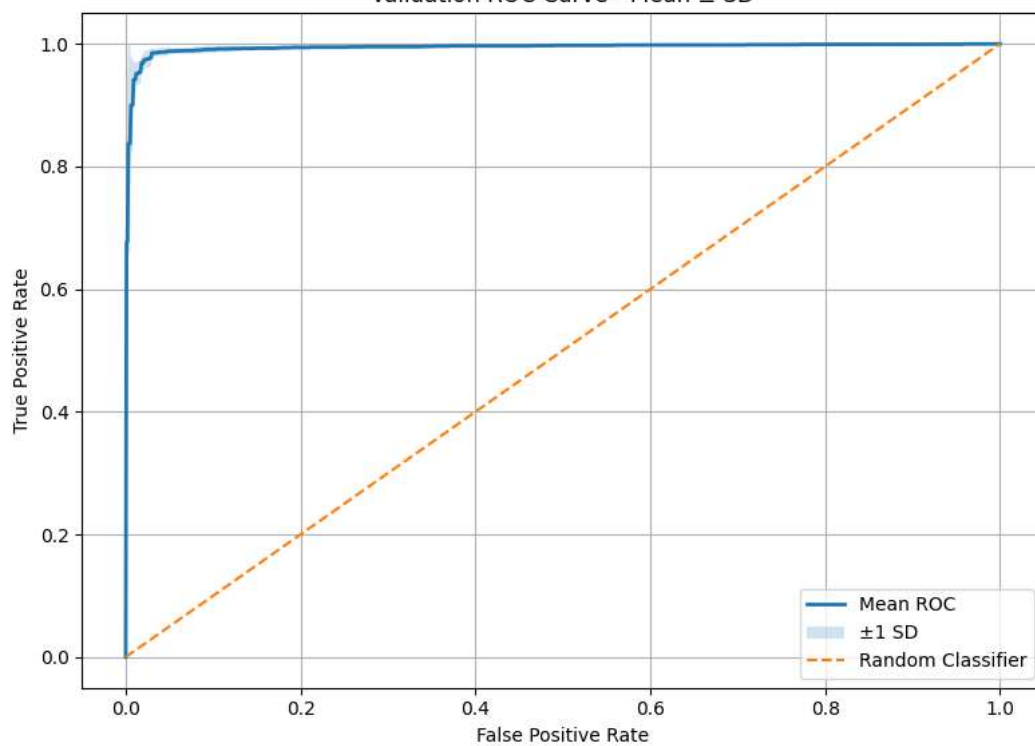


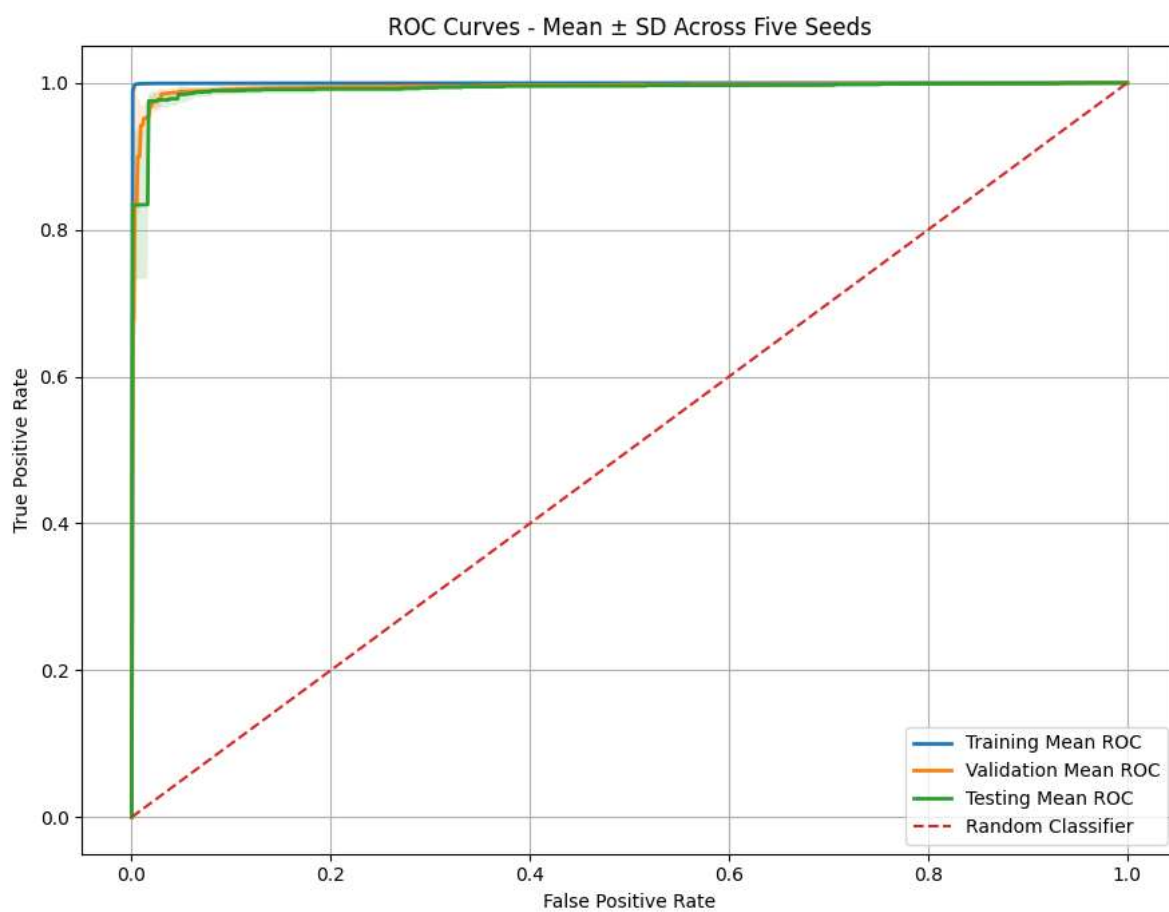
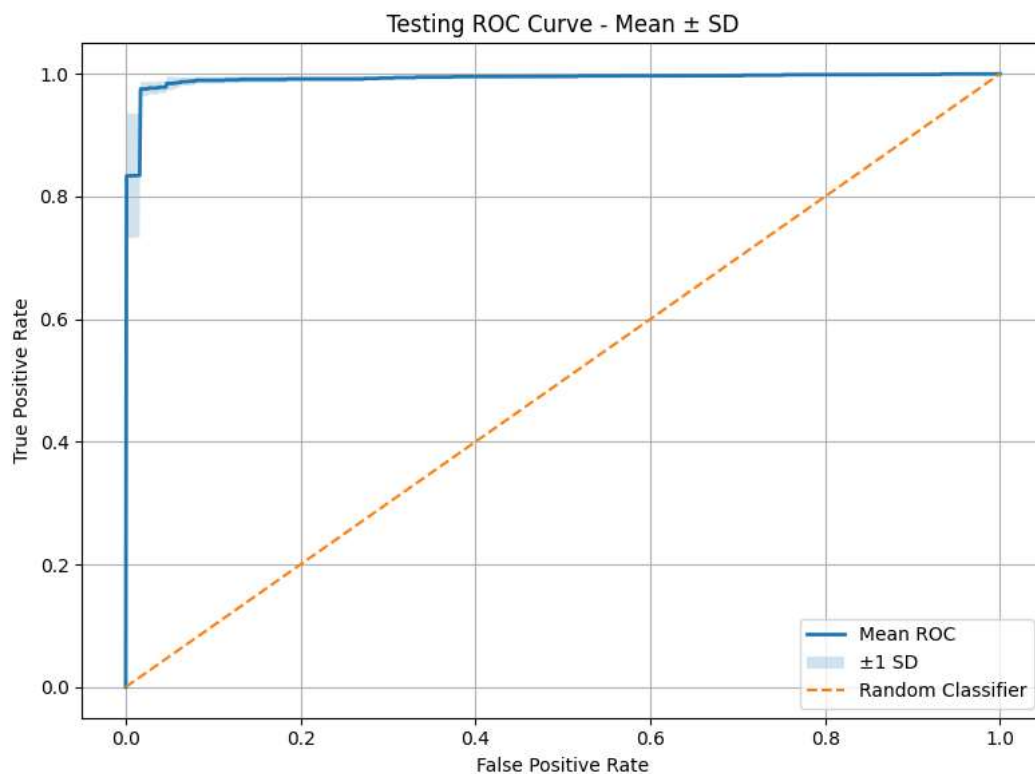
Individual ROC Curves - Validation

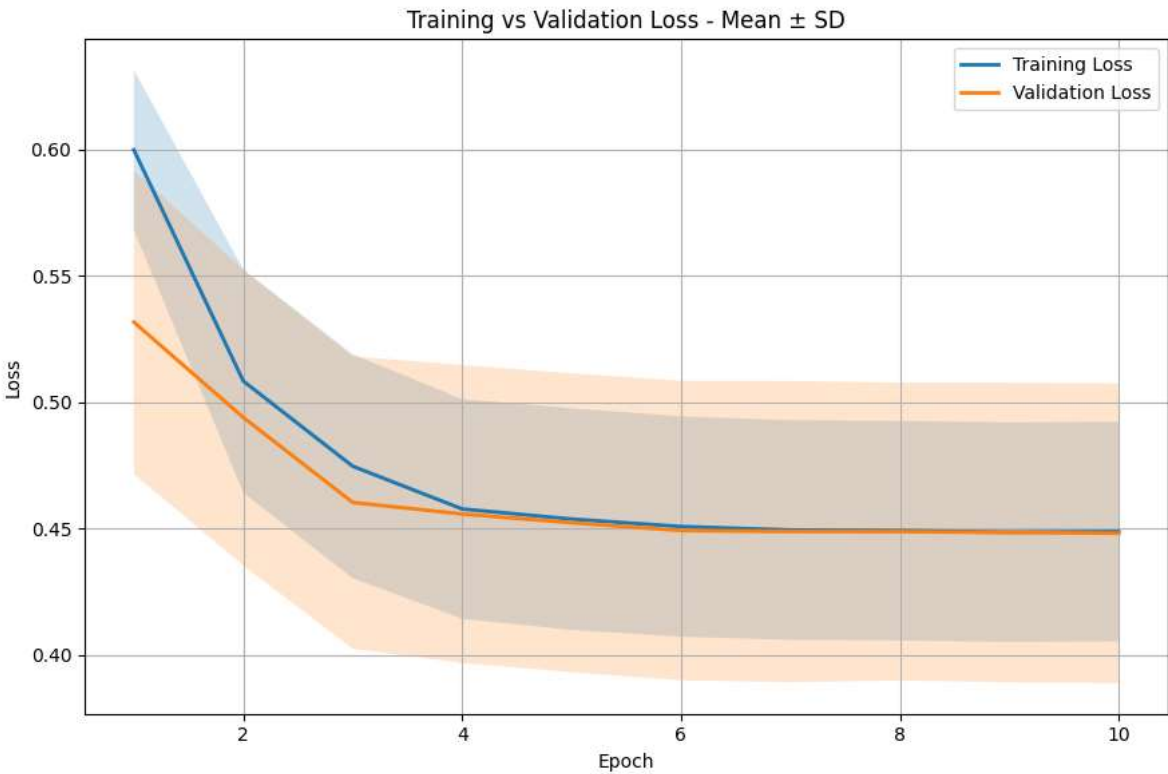
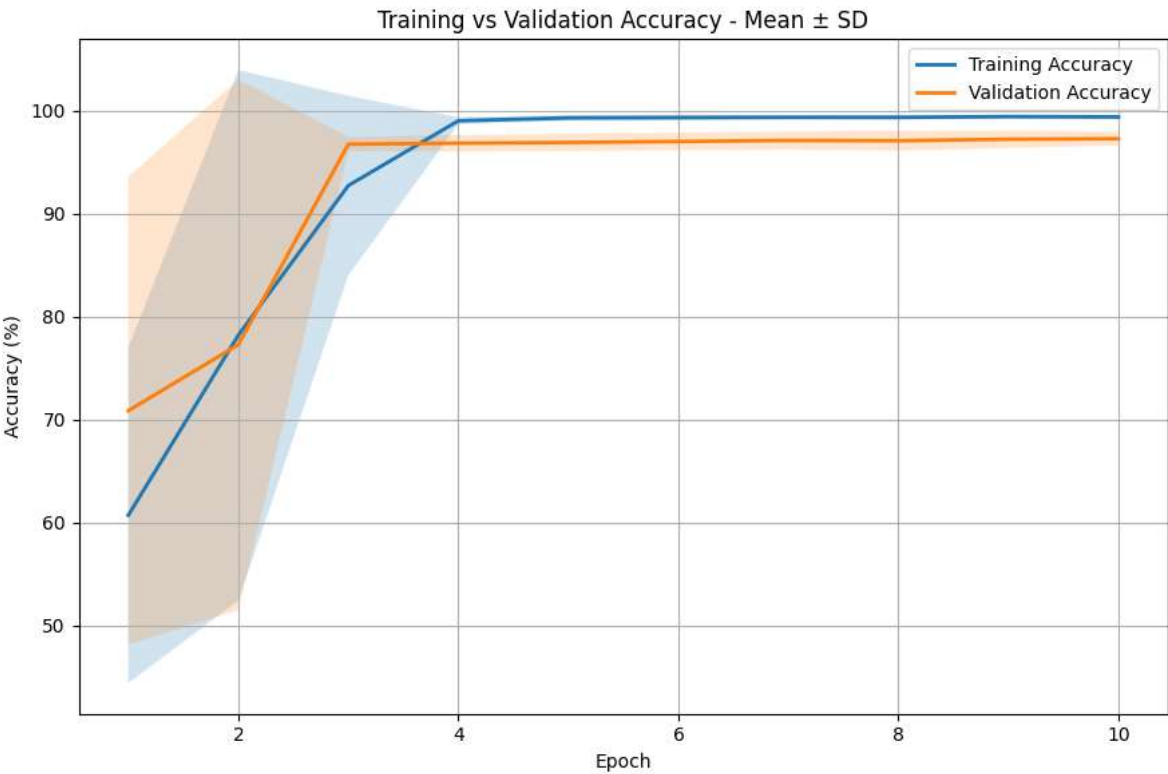


Individual ROC Curves - Testing



Training ROC Curve - Mean $\pm$ SDValidation ROC Curve - Mean $\pm$ SD





=====

TIME FOR EACH SEED

=====

Seed	Time (minutes)	Time (seconds)
42	74.727795	4483.667693
52	12.371408	742.284467
62	12.689211	761.352645
72	12.924423	775.465377
82	12.481505	748.890330

=====

MEAN $\pm$ SD TIME

=====

Mean Time: 25.04 minutes

SD Time: 27.78 minutes

Mean $\pm$ SD Time: 25.04 $\pm$ 27.78 minutes

=====

FINAL FIVE-SEED EXPERIMENT COMPLETED

=====

Architecture:

EfficientNet-B0 $\rightarrow$ 1280 $\rightarrow$ Linear(1280 $\rightarrow$ 4) $\rightarrow$ Tanh $\rightarrow$ Linear(4 $\rightarrow$ 5) $\rightarrow$ ReLU $\rightarrow$ Linear(5 $\rightarrow$ 1) $\rightarrow$ Sigmoid

Seeds:

[42, 52, 62, 72, 82]

All seeds completed:

[42, 52, 62, 72, 82]

Results directory:

/content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results

Individual results:

/content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/individual_five_seed_results.csv

Mean $\pm$ SD results:

/content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/mean_sd_performance.csv

Timing:

/content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/five_seed_timing.csv

Figures:

/content/drive/MyDrive/Bone_Fracture_Binary_Classification/EfficientNet_NonlinearMLP_5Seed_Results/figures

Mean $\pm$ SD Time:

25.04 $\pm$ 27.78 minutes

=====

DONE

=====